Constraint Projection in Discrete Ledger Systems: A Formal Model of Future-Balance Visibility with Sleep-Stage Predictions

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Abstract

We present a formal model in which future constraints become visible in the present without retrocausation. The model operates within Recognition Science (RS), a discrete-time framework whose axioms—a unique cost function, a forced golden-ratio scale, three spatial dimensions, and an 8-tick conservation period—have been published and machine-verified elsewhere [20, 21, 24, 22]. Taking these axioms as given, we prove that any observation event within an 8-tick window creates a *balance debt* that the remaining ticks must discharge (Theorem 3.1), and that this debt inflates the effective cost function monotonically (Theorem 3.7). We call this cost inflation "Phantom Light"—the shadow of a future balance requirement visible in the present as a restriction on admissible trajectories.

We then model sleep stages by a monotone coherence ordering (an empirical input from polysomnography) and prove a robust visibility ordering: lower-coherence stages carry higher phantom magnitude, with N3 dominating N1 and REM (Theorem 4.4). Under the canonical consecutive-integer budget used for illustration, N3 has twice the N1 phantom magnitude. Zero balance debt annihilates phantom light in every mode (Theorem 4.8).

The model yields five falsifiable predictions, including a sleep-stage ordering of precognitive dream reports and a presentiment-scaling law. We also specify preregistered empirical proxies for the latent model variables, a blinded prospective dream-scoring protocol, and a small-effect power envelope for confirmatory studies. All theorems are machine-verified in the Lean 4 proof assistant (52 theorems, zero unproved placeholders, zero new axioms beyond the RS foundation). Each claim is classified as PROVED, MODELING CHOICE, or HYPOTHESIS in an explicit table (§9). We discuss the relationship to Aharonov's Two-State Vector Formalism, to Hobson's AIM model of dream generation, and to the presentiment literature including both positive reports (Radin, Mossbridge) and critical responses (Wagenmakers, Alcock).

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1 Introduction

1.1 Motivation

Standard physics evolves systems forward from initial conditions. Yet several theoretical frameworks in quantum foundations incorporate future boundary conditions: Hamilton's Principle of Least Action selects paths by endpoint constraints, Feynman's path integral weights amplitudes by boundary connectivity, and Aharonov, Bergmann, and Lebowitz's Two-State Vector Formalism (TSVF) describes quantum systems using both pre-selected and post-selected states [1, 2]. In each case, the "future" influences the present not through retrocausal signaling but through constraint on the space of admissible trajectories.

This paper develops a concrete, finite-dimensional model of such constraint projection within Recognition Science (RS), a discrete-time framework that enforces conservation over sliding 8-tick windows. The key insight is simple: when a recognition event (an observation) pins one slot of the window to a nonzero value, the remaining slots are *mathematically obligated* to compensate. This obligation—the "balance debt"—inflates the effective cost of all configurations that would make compensation difficult. We formalize this inflation, prove its mathematical properties, and derive testable predictions about when and how the phantom field becomes detectable.

1.2 Why Consciousness Studies?

The model's predictions concern consciousness because they are *about states of reduced coherence*. During waking, the high coherence of neural processing suppresses phantom visibility. During sleep, coherence drops stage by stage (a well-established finding from polysomnography [10, 30]), and the model predicts a corresponding increase in sensitivity to future constraints. This connects directly to the long-standing observation that dreams—particularly those occurring in deep sleep—sometimes appear to contain information about future events [18, 28, 12].

We are careful to distinguish what is proved from what is hypothesized. The mathematical structure (balance debt, cost inflation, sleep-stage ordering) is machine-verified. The claim that biological systems can *detect* phantom fields is an empirical hypothesis for which we provide falsification protocols. The paper engages with both the positive presentiment literature [17, 16] and its methodological critiques [29, 3].

1.3 Contributions

1. A formal model of future-constraint visibility based on windowed conservation (§3), with all theorems machine-verified.

2. A sleep-stage hierarchy of phantom visibility (§4), linking the model to empirical sleep architecture.

3. A preregistered empirical bridge from latent RS variables to dream and presentiment observables, including a blinded prospective dream-scoring protocol (§7.2–§7.3).

4. Explicit falsification criteria (§7) with quantitative thresholds and small-effect design targets.

5. A claim hygiene table (§9) classifying every statement as PROVED, MODEL, or HYPOTHESIS.

2 Axioms and Notation

We work within the Recognition Science (RS) framework, whose foundations have been developed in a series of publications [20, 21, 24, 22, 23]. The present paper does not re-derive the RS axioms; it takes them as given and derives consequences for consciousness and sleep. For the reader's convenience, we state the axioms used in this paper. Each has been machine-verified in Lean 4 with zero unproved placeholders [25].

Axiom 1 (Cost Function). *There exists a unique cost function $J : \mathbb{R}_{>0} \to \mathbb{R}_{\geq 0}$ satisfying the Recognition Composition Law $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ with $J(1) = 0$ and $J''_{\log}(0) = 1$. The unique solution is $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.* [T5; Refs. [21, 22]]

Axiom 2 (Golden Ratio). *Self-similarity in the discrete ledger forces the scale ratio $\varphi = (1 + \sqrt{5})/2$ as the unique positive root of $x^2 = x + 1$.* [T6; Ref. [20]]

Axiom 3 (Dimension). *Non-trivial linking requires $D = 3$ spatial dimensions.* [T8; Ref. [24]]

Axiom 4 (8-Tick Period). *The minimal ledger-compatible walk on the 3-cube Q_3 has period $2^3 = 8$, realized by a Gray code Hamiltonian cycle.* [T7; Ref. [20]]

Axiom 5 (Window Neutrality). *Every aligned 8-tick window satisfies:*

$$\sum_{k=0}^{7} s(t_0 + k) = 0 \qquad \forall\, t_0 \equiv 0 \pmod 8. \tag{1}$$

This is the double-entry conservation law applied to time. [Ref. [20]; Lean: `LNAL.Invariants.neutral_every_8th_`

Axiom 6 (Cost Minimization). *The recognition operator $\hat{R}$ evolves the system state over 8 ticks, selecting the trajectory that minimizes integrated J-cost subject to the neutrality constraint.* [Ref. [20]]

2.1 Notation

Symbol	Meaning
$w : \{0, \ldots, 7\} \to \mathbb{Z}$	Window signal
δ	Contribution of a recognition event
$\mathcal{D}$	Balance debt (running sum within window)
R	Remaining ticks until window closure
Φ_{mag}	Phantom magnitude
$C(\text{mode})$	Coherence budget for a sleep stage
Θ	Global phase (if GCIC holds)

2.2 Formal Verification

All theorems in §3–§6 are verified in Lean 4 with the Mathlib library. The relevant modules (52 theorems total) contain zero `sorry` placeholders and introduce zero new axioms beyond the RS foundation. Build instructions and a complete theorem table are provided in the appendix. The Lean source is available at [25].

3 Phantom Light: Balance Debt and Cost Inflation

3.1 Recognition Events and Balance Debt

A *recognition event* at position $p \in \{0, \ldots, 7\}$ with contribution $\delta \in \mathbb{Z} \setminus \{0\}$ records a value in the current window slot. (In the RS formalism, this is called a LOCK event.) As events accumulate, the window develops a *balance debt*: $\mathcal{D} = \sum_{j \leq k} w(j)$.

Theorem 3.1 (Recognition Forces Future Balance)**.** *If a recognition event with contribution δ occurs at position p in a neutral window, the sum of all other contributions equals $-\delta$:*

$$\sum_{k \neq p} w(k) = -\delta.$$

Proof sketch. By neutrality (Axiom 5), $\sum_k w(k) = 0$. Decompose: $\delta + \sum_{k \neq p} w(k) = 0$. Rearrange. [Lean: `lock_forces_future_balance`] □

This is the foundational result. A recognition event is not an isolated act: it creates a debt that the future is mathematically obligated to repay within the current window.

3.2 Phantom Magnitude

We quantify the "visibility" of the future constraint:

Definition 3.2 (Phantom Magnitude)**.** For balance debt $\mathcal{D} \in \mathbb{Z}$ and remaining ticks $R \geq 0$:

$$\Phi_{\text{mag}}(\mathcal{D}, R) := \frac{|\mathcal{D}|}{R+1}. \tag{2}$$

Theorem 3.3 (Phantom Properties)**.** (a) $\Phi_{\text{mag}} \geq 0$.

(b) *Monotone increasing in $|\mathcal{D}|$ for fixed R.*

(c) *Monotone decreasing in R for fixed $\mathcal{D} \neq 0$.*

(d) $\Phi_{\text{mag}}(0, R) = 0$*: zero debt implies zero phantom.*

(e) $\Phi_{\text{mag}}(\mathcal{D}, 0) = |\mathcal{D}|$*: at the boundary, the full debt is exposed.*
[Lean: `phiMag_nonneg`, `phiMag_mono_abs_debt`, `phiMag_anti_mono_R`, `phiMag_zero_debt`]

Property (c) is particularly important: as the window closes, the phantom grows more urgent. This creates a natural "deadline effect" that intensifies the future constraint.

3.3 Backward Projection Principle

Theorem 3.4 (Backward Projection)**.** *If each remaining position is bounded in magnitude by $B \geq 0$, then any feasible completion satisfying neutrality requires $|\mathcal{D}| \leq B \cdot R$. Equivalently: if $|\mathcal{D}| > BR$, no feasible completion exists.* [Lean: `backward_projection_principle`]

Corollary 3.5. $\Phi_{\text{mag}} \leq BR/(R+1) < B$. *The phantom never exceeds the per-tick capacity. The precognitive horizon is finite and bounded.*

3.4 Cost Inflation

Definition 3.6 (Augmented Cost)**.** For configuration $x > 0$, phantom state $(\mathcal{D}, R)$, and penalty scale $\lambda > 0$:

$$J_{\text{PL}}(x; \mathcal{D}, R) := J(x) + \lambda \cdot \Phi_{\text{mag}}(\mathcal{D}, R). \tag{3}$$

Theorem 3.7 (Phantom Inflates Cost)**.** $J_{\text{PL}}(x; \mathcal{D}, R) \geq J(x)$ *for all $x > 0$ when $\lambda > 0$. Equality holds if and only if $\mathcal{D} = 0$.* [Lean: `phantom_inflates_cost`; `zero_phantom_pure_cost`]

Since $\hat{R}$ minimizes J-cost (Axiom 6), trajectories that incur high debt without rapid repayment are statistically suppressed. This is the mechanism by which future constraints bias the present.

3.5 Boundary Reset

Theorem 3.8 (Phantom Vanishes at Window Boundaries)**.** *At every aligned 8-tick boundary (tick $8k$ for $k \geq 0$), the phantom magnitude is exactly zero.* [Lean: `phantom_zero_at_lnal_boundaries`]

The phantom field is not permanent. Each window closure resets the slate—a "clean start" at every 8-tick boundary.

3.6 Constraint Projection, Not Retrocausation

It is essential to clarify the ontological status of Phantom Light. It is *not* a signal traveling backward in time. No information violates causality. Instead, Phantom Light represents a restriction on the set of admissible trajectories:

> *Because the window must close to zero, the set of allowed paths from the current state is a strict subset of all combinatorially possible paths. This reduction in phase-space volume is information accessible in the present.*

This is conceptually identical to the mechanism underlying variational principles in classical mechanics: a particle following Hamilton's principle "knows" its endpoint because the set of stationary-action paths is constrained by boundary conditions at both ends. In quantum mechanics, Aharonov's TSVF makes this explicit by describing states with both forward and backward evolving state vectors [2]. The distinctive contribution of the present model is a *discrete, finite* version of this structure: the precognitive horizon is exactly ≤ 7 ticks (within one window), and the constraint strength Φ_{mag} is a computable function of the debt and remaining time.

4 Sleep Architecture and Phantom Visibility

4.1 Empirical Background: Sleep Stages and Coherence

The classification of sleep into distinct stages is one of the most robust findings in sleep science. Polysomnographic recordings reveal a progression from waking through light sleep (N1, N2) to deep slow-wave sleep (N3) and REM, characterized by progressive decreases in cortical coherence and high-frequency EEG power [10, 30, 27]. EEG synchronization decreases monotonically from waking through N3, reflecting reduced thalamocortical connectivity [14].

In Hobson's Activation-Input-Modulation (AIM) model [9], the "activation" dimension A tracks cortical arousal, which decreases from waking to deep sleep. Our coherence budget C corresponds to the A dimension: a monotone ordering that reflects the empirically measured progression.

4.2 The Coherence Budget Model

Definition 4.1 (Coherence Budget)**.** Each sleep mode is assigned a non-negative integer coherence budget with the ordering:

$$C_{\text{awake}} > C_{\text{REM}} > C_{\text{N1}} > C_{\text{N2}} > C_{\text{N3}}.$$

Remark 4.2 (Status: Modeling Choice). The specific integers (5, 4, 3, 2, 1 in our formalization) are a modeling choice. The theorems below depend only on the *ordering*, which is an empirical fact of sleep science. Any monotone decreasing assignment $C : \text{SleepMode} \to \mathbb{N}$ with the same ordering produces identical qualitative results. We select consecutive integers as the simplest choice consistent with the data.

4.3 Sleep-Stage Phantom Magnitude

The sleep-stage phantom magnitude uses the coherence budget as the denominator:

$$\Phi_{\text{sleep}}(\text{mode}, \mathcal{D}) := \frac{|\mathcal{D}|}{C(\text{mode}) + 1}. \tag{4}$$

The intuition is that higher cortical coherence acts as "noise" that masks the phantom signal. During deep sleep, when coherence is minimal, the phantom field becomes maximally visible—analogous to how faint stars become visible only when the sky is dark.

Theorem 4.3 (Lower Coherence Increases Visibility)**.** *If $C(m_1) \leq C(m_2)$, then $\Phi_{\text{sleep}}(m_2, \mathcal{D}) \leq \Phi_{\text{sleep}}(m_1, \mathcal{D})$ for all $\mathcal{D}$.* [Lean: `lower_coherence_increases_phantom_visibility`]

4.4 The Sleep-Stage Hierarchy

Theorem 4.4 (Deep Sleep Dominates)**.** *For all balance debt $\mathcal{D}$:*

$$\Phi_{\text{sleep}}(\text{N1}, \mathcal{D}) \leq \Phi_{\text{sleep}}(\text{N3}, \mathcal{D}), \tag{5}$$

$$\Phi_{\text{sleep}}(\text{REM}, \mathcal{D}) \leq \Phi_{\text{sleep}}(\text{N3}, \mathcal{D}). \tag{6}$$

Under the canonical consecutive-integer coherence budget used in this paper, deep sleep has exactly twice the phantom magnitude of light sleep: $\Phi_{\text{sleep}}(\text{N3}) = 2 \cdot \Phi_{\text{sleep}}(\text{N1})$.
[Lean: `stage3_phantom_visibility_ge_stage1`; `stage3_phantom_visibility_ge_rem`; `phantomSleepMagnitude_rat`

Remark 4.5. The exact factor of 2 depends on the specific budget integers (1 and 3 for N3 and N1). With the consecutive-integer choice, the ratio is $\frac{|\mathcal{D}|/2}{|\mathcal{D}|/4} = 2$. A different monotone assignment would yield a different ratio, but the *ordering* N3 $\geq$ N1 $\geq$ REM is invariant under any assignment consistent with the empirical coherence ordering.

4.5 Explicit Formulas and Bounds

Theorem 4.6 (Formulas and Bounds)**.** *$\Phi_{\text{sleep}}(\text{awake}, \mathcal{D}) = |\mathcal{D}|/6$ and $\Phi_{\text{sleep}}(\text{N3}, \mathcal{D}) = |\mathcal{D}|/2$. For all modes: $|\mathcal{D}|/6 \leq \Phi_{\text{sleep}} \leq |\mathcal{D}|/2$.*
[Lean: `awake_phantom_formula`; `stage3_phantom_formula`; `phantomSleepMagnitude_bounds`]

Deep sleep provides three times the phantom visibility of waking.

4.6 Stage-Weighted Exposure

Each sleep stage has a characteristic time window $W(\text{mode})$. The total phantom exposure is $E(\text{mode}, \mathcal{D}) = W(\text{mode}) \cdot \Phi_{\text{sleep}}(\text{mode}, \mathcal{D})$.

Theorem 4.7 (Deep Sleep Exposure Dominates)**.** *$E(\text{N1}) \leq E(\text{N3})$ and $E(\text{REM}) \leq E(\text{N3})$.*
[Lean: `stage3_exposure_ge_stage1`; `stage3_exposure_ge_rem`]

N3 sleep dominates in both *per-tick visibility* and *total exposure*. This is the formal basis for the prediction that deep-sleep dreams carry the strongest precognitive content.

4.7 Zero Debt: Dreamless Waking

Theorem 4.8 (Zero Debt Eliminates Sleep Phantom)**.** *If balance debt is zero, phantom magnitude vanishes in every sleep mode: $\Phi_{\text{sleep}}(\text{mode}, 0) = 0$ for all modes.*
[Lean: `zero_debt_gives_zero_phantom_sleep_magnitude`]

A system with no outstanding ledger obligations has no phantom field to sense. The theorem concerns latent balance debt, not a directly observable psychological variable. Empirically, we test it through the preregistered future-event load proxies defined in §7.2: nights followed by very low future-event load should yield fewer above-chance future matches.

5 Ultradian Architecture

The 90-minute Basic Rest-Activity Cycle (BRAC) is one of the most robust findings in sleep science [11, 4].

Theorem 5.1 (Ultradian Count)**.** $\lfloor 1440/90 \rfloor = 16$ *ultradian cycles tile each 24-hour day.*
[Lean: `sixteen_ultradians`]

Remark 5.2 (Status: Empirical Input)*.* The 90-minute BRAC duration is an empirical input, not derived from RS axioms. We record it as a constant and prove only arithmetic consequences. We note that $90 = F_{11} + 1$ (where $F_{11} = 89$ is the 11th Fibonacci number), which is structurally consistent with the φ-ladder architecture of RS but is not claimed as a derivation.

6 Consciousness Interface

6.1 Phantom Sensing

We model the detection of phantom fields by a boundary (a localized coherent subsystem in RS) through a sensitivity parameter $\sigma > 0$. The perceived phantom signal is:

$$\text{Perceived} = \sigma \cdot \Phi_{\text{mag}}.$$

Theorem 6.1 (Positivity of Phantom Readout)**.** *If* $\sigma > 0$ *and* $\mathcal{D} \neq 0$*, then* Perceived > 0.
[Lean: `precognition_via_phantom_sensing`]

Remark 6.2. This is an algebraic tautology: a positive number times a positive number is positive. Its value lies not in its depth but in its *precision*: it states exactly what is needed for a biological system to detect phantom fields. Whether $\sigma > 0$ for human neural systems is the central empirical question, addressed in §7.

6.2 Response Time

Theorem 6.3 (Response Time Law)**.** *For any phantom sensing configuration, there exists a positive response time* $\tau = 1/(\Phi_{\text{mag}} + 1) > 0$*. Larger phantom magnitudes yield shorter response times.*
[Lean: `precognition_response_time_law`]

This predicts that high-stakes future events should be precognized *earlier* than low-stakes events—a quantitative relationship between event magnitude and anticipation time.

6.3 Global Phase Coupling (Conditional)

Theorem 6.4 (Θ-Coupling Bound)**.** *If the Global Co-Identity Constraint (GCIC) holds—i.e., all coherent subsystems share a single universal phase* Θ*—then for any pair of subsystems, there exists a coupling coefficient* $C = \cos(2\pi \cdot \Delta\Theta)$ *with* $|C| \leq 1$.
[Lean: `remote_viewing_via_theta`]

Remark 6.5 (Conditional on GCIC)*.* GCIC is an internal hypothesis of RS, not a consequence of the axioms listed in §2. All claims about non-local phantom sensing (remote viewing, telepathy) are *strictly conditional* on GCIC. This paper does not argue for or against GCIC; it derives consequences conditional on it and provides falsification protocols.

7 Predictions and Falsification

7.1 Empirical Bridge

The following predictions require only the RS axioms (§2), the proved theorems (§3–§6), the empirical coherence ordering, and the single assumption that biological neural systems have positive phantom sensitivity ($\sigma > 0$).

7.2 Operationalization for Dream and Presentiment Studies

The formal variables $\mathcal{D}$, R, and $\Phi_{\text{mag}}(\mathcal{D}, R)$ are latent RS quantities. They are not directly observed in human experiments. For empirical work we therefore introduce preregistered proxies designed to preserve the two monotonicities proved in Theorem 3.3: larger debt should increase phantom strength, and shorter temporal distance should increase urgency.

Dream studies. For a night n and horizon H days, let $E_n(H)$ be the set of salient events that actually occur to the participant during the following horizon. After the horizon closes and blinded to dream text, each event $e \in E_n(H)$ receives an impact score $I_e \in \{0, 1, 2, 3\}$ from a preregistered rubric and a temporal distance $d_e \in \{0, \ldots, H-1\}$ measured in days from the dream report. We then define the future-event load proxy

$$\widehat{\mathcal{D}}_n(H) := \sum_{e \in E_n(H)} \frac{I_e}{d_e + 1}. \tag{7}$$

This is not the latent ledger debt itself; it is an observable proxy that tracks the model's two key monotones: event impact and temporal proximity.

Each dream report should be time-stamped immediately on awakening, locked before the future interval opens, and scored only after the horizon closes. In the confirmatory design, each report is presented to independent judges together with one true future-event packet and three decoy packets matched for category and base rate. Judges, blind to sleep stage, chronology, and packet identity, rank the packets by match quality using a preregistered rubric. The primary endpoint is top-rank accuracy above the 25% chance baseline; secondary endpoints include mean reciprocal rank and continuous similarity scores. Inter-rater reliability should reach Fleiss' $\kappa \geq 0.70$ before confirmatory scoring proceeds.

Presentiment studies. For trial t, let $a_t \in \{1, 2, 3\}$ denote preregistered low/medium/high stimulus intensity assigned by hardware RNG, and let r_t denote the number of preregistered pre-stimulus analysis bins remaining until onset. Define the empirical proxies

$$\widehat{\mathcal{D}}_t := a_t, \qquad \widehat{\Phi}_t := \frac{a_t}{r_t + 1}. \tag{8}$$

For a preregistered physiological summary statistic Y_t (for example, baseline-corrected skin conductance or pupil dilation), the confirmatory question is whether the regression slope in $Y_t = \alpha + \beta \widehat{\Phi}_t + \varepsilon_t$ is positive.

7.3 Model-Based Effect-Size Envelope

The theorems fix monotonicity, ordering, and certain canonical ratios, but they do not fix absolute human effect sizes because the biological sensitivity parameter σ is not yet known. We therefore separate *scale-free* consequences from *study-design* amplitude expectations.

If an empirical readout is locally linear in perceived phantom magnitude over the tested range, then

$$\mathbb{E}[Y] = \alpha + \beta \widehat{\Phi}, \qquad \beta > 0. \tag{9}$$

This yields three scale-free expectations:

1. At fixed temporal lag, low/medium/high stimulus bins coded as $(1, 2, 3)$ should show approximately a $1 : 2 : 3$ response ordering.
2. At fixed stimulus intensity, adjacent pre-stimulus bins should grow by the factor $(r + 2)/(r + 1)$ as onset approaches.
3. In sleep data, N3 should exceed N1 and REM for any monotone coherence budget; the exact $2 : 1$ N3:N1 ratio is expected only under the canonical consecutive-integer budget used here.

Because ordinary waking experience does not reveal obvious precognitive perception, any human value of σ must be small at the population level if it exists at all. Confirmatory studies should therefore be powered for small aggregate effects rather than dramatic ones. As a pragmatic design envelope consistent with the existing presentiment meta-analytic literature [16], we recommend preregistering physiological targets in the low-single-digit correlation range (roughly $|r| = 0.01$–0.05) or standardized mean differences below $d = 0.2$. In blinded four-choice dream matching, the practically relevant signal is likewise expected to appear as a modest lift above the 25% chance baseline rather than near-perfect classification. These numerical bands are study-design heuristics, not proved theorems.

7.4 Predictions

Prediction 7.1 (Deep-Sleep Precognitive Dominance)**.** In a prospective blinded rank-matching design, reports collected after polysomnography-confirmed N3 awakenings should achieve higher top-rank accuracy and mean reciprocal rank than reports collected after N1 or REM awakenings, by Theorem 4.4.

Prediction 7.2 (Low Debt, Low Match Rate)**.** Nights with lower preregistered future-event load proxy $\widehat{\mathcal{D}}_n(H)$ should yield fewer above-chance future-event matches than nights with higher $\widehat{\mathcal{D}}_n(H)$, by Theorem 4.8.

Prediction 7.3 (Debt-Content Correlation)**.** Within the blinded future-event packets, higher-impact and more proximate events should receive higher match scores than lower-impact or more distant events, consistent with the proxy $\widehat{\mathcal{D}}_n(H)$ and Theorem 3.3(b).

Prediction 7.4 (Response Time Scaling)**.** Pre-stimulus physiological response should show a positive slope against the preregistered proxy $\widehat{\Phi}_t = a_t/(r_t + 1)$, with earlier and larger responses for higher-intensity stimuli, by Theorem 6.3.

Prediction 7.5 (Boundary Reset)**.** The presentiment effect, if it exists, should show periodic attenuation at intervals corresponding to 8-tick boundaries, by Theorem 3.8.

7.5 Falsification Criteria

Falsification Criterion 7.6 (No Precognition Effect)**.** If $N > 10{,}000$ trials yield a 95% confidence interval excluding positive correlations larger than $r = 0.01$ between pre-stimulus physiology and the preregistered proxy $\widehat{\Phi}_t$, the phantom sensing hypothesis ($\sigma > 0$) is falsified at practically relevant human effect sizes.

Falsification Criterion 7.7 (No Sleep-Stage Effect)**.** If N3 dream reports are no more accurate than N1 or REM reports on the preregistered primary endpoint across $N > 200$ subjects with polysomnographic staging, the sleep-stage visibility hierarchy is falsified.

Falsification Criterion 7.8 (Uniform Presentiment)**.** If presentiment magnitude is statistically indistinguishable across stimulus intensities ($p > 0.05$, $BF_{01} > 3$), the debt-scaling model is falsified.

7.6 Experimental Design Principles

Any experimental test must adhere to:

1. **Pre-registration**: Full protocol registered on OSF or AsPredicted before data collection.
2. **Time-locking**: Dream reports must be recorded, time-stamped, and locked before the future interval opens.
3. **Polysomnographic staging**: Sleep-stage labels must be assigned from preregistered scoring rules by raters blind to dream content and outcome.
4. **Double-blinding**: Experimenters interacting with subjects, dream judges, and data analysts must be blind to condition, stimulus schedule, and target identity until lock.
5. **Matched decoys**: Each dream must be scored against one true future-event packet and decoy packets matched for category and base rate.
6. **Judge reliability**: Inter-rater reliability on the primary dream-scoring rubric should satisfy Fleiss' $\kappa \geq 0.70$ before confirmatory unblinding.
7. **Hardware RNG**: Future events determined by quantum or thermal random number generators with verified entropy.
8. **Fixed primary endpoints**: Primary outcome measures (top-rank accuracy, mean reciprocal rank, regression slope, latency window) must be designated in advance.
9. **Multiple-comparison correction**: Bonferroni or FDR (Benjamini-Hochberg) for multiple physiological channels.
10. **Bayesian analysis**: Report both frequentist p-values and Bayes factors (BF_{10}) to distinguish "evidence for null" from "inconclusive."
11. **Permutation controls**: Shuffle time labels and packet identities to generate null distributions.

These standards are drawn from the methodological critiques of Wagenmakers et al. [29], who argued that prior presentiment studies suffered from optional stopping, flexible analysis, and insufficient reporting of null results.

8 Related Work

8.1 Two-Time Boundary Conditions in Physics

The idea that future boundary conditions constrain present dynamics has a long history. Hamilton's Principle of Least Action selects trajectories by minimizing the action integral over a fixed time interval with fixed endpoints [13]. Feynman's path integral sums amplitudes over all paths connecting initial and final states. Most directly relevant is Aharonov's TSVF [1, 2], which describes quantum systems using both a forward-evolving state (from preparation) and a backward-evolving state (from post-selection). The present model is a discrete, finite-dimensional analog: the window neutrality constraint plays the role of post-selection, and the phantom magnitude quantifies how strongly the post-selection constrains the present.

The Wheeler-Feynman absorber theory [31] similarly requires advanced (backward-in-time) solutions for consistency, though these are usually interpreted as canceling in the classical limit. Cramer's Transactional Interpretation [7] extends this to a full interpretation of quantum mechanics.

8.2 Consciousness and Time

The relationship between consciousness and temporal experience has been explored from multiple directions. Tononi's Integrated Information Theory (IIT) [26] proposes that consciousness corresponds to integrated information (Φ), which depends on the causal structure of a system over time. The "temporal depth" of Φ in IIT bears a structural resemblance to our phantom magnitude: both quantify how much the past-future structure of a system matters for its present state.

Global Workspace Theory (GWT) [5, 8] models consciousness as a broadcast mechanism. In GWT terms, phantom light would correspond to a constraint-derived signal that becomes available for global broadcast during reduced-coherence states (sleep), when the usual sensory competition for the workspace is diminished.

8.3 Dreams and Precognition

The claim that dreams sometimes contain future information has a long history in both anecdotal reports and experimental investigation. Rhine [18] and Van de Castle [28] compiled systematic case collections. Kramer [12] reviewed the dream-cognition literature from a psychiatric perspective.

Hobson's AIM model [9, 10] explains dream phenomenology through three dimensions: Activation, Input gating, and chemical Modulation. In AIM terms, our coherence budget C corresponds to the Activation dimension, and the prediction that deep-sleep dreams carry stronger phantom content maps to the low-A region of AIM state space.

8.4 The Presentiment Literature

Presentiment (physiological anticipation of random future stimuli) has been reported by Radin [17], Bierman and Radin [6], McCraty et al. [15], and reviewed by Mossbridge et al. [16], who conducted a meta-analysis of 26 studies and reported a small but statistically significant anticipatory effect.

However, this literature has been subject to serious methodological critique. Wagenmakers et al. [29] argued that Bem's precognition results were artifacts of optional stopping and flexible analysis. Alcock [3] reviewed the broader history of parapsychological claims and argued that replication failures are systematic. Rouder and Morey [19] reanalyzed Bem's data using Bayesian methods and found the evidence equivocal.

Our model does not assume the presentiment literature is correct. It provides a framework within which presentiment, *if it exists*, would have specific properties (sleep-stage hierarchy, debt scaling, response time law) that are not predicted by artifact explanations. The falsification criteria in §7 are designed to distinguish genuine phantom sensing from the methodological artifacts identified by critics.

9 Claim Hygiene

Every claim in this paper is classified below. "PROVED" means machine-verified from the RS axioms together with any explicitly declared modeling assumptions used in the statement. "MODEL" means a definition, operationalization, or parameter choice that is consistent with empirical data but not itself derived from axioms. "HYPOTHESIS" means a falsifiable empirical claim.

Claim	**Status**	**Reference**
RS Axioms (taken as given)		

Claim	Status	Reference
J unique (T5)	AXIOM	[21]
φ forced (T6)	AXIOM	[20]
$D = 3$ (T8)	AXIOM	[24]
8-tick period (T7)	AXIOM	[20]
Window neutrality	AXIOM	[20]
Cost minimization	AXIOM	[20]
Phantom Light (proved from axioms)		
Recognition forces balance	PROVED	Thm. 3.1
$\Phi_{\text{mag}} \geq 0$	PROVED	Thm. 3.3
Monotone in $\lvert\mathcal{D}\rvert$	PROVED	Thm. 3.3
Anti-monotone in R	PROVED	Thm. 3.3
Zero debt = zero phantom	PROVED	Thm. 3.3
Backward projection	PROVED	Thm. 3.4
Cost inflation	PROVED	Thm. 3.7
Boundary reset	PROVED	Thm. 3.8
Sleep model (proved from axioms + ordering)		
Coherence ordering	MODEL	Def. 4.1
Canonical integer budget	MODEL	Rem. after Thm. 4.4
Stage window durations	MODEL	empirical
N3 $\geq$ N1 visibility	PROVED	Thm. 4.4
N3 $\geq$ REM visibility	PROVED	Thm. 4.4
Canonical budget: N3 = 2$\times$ N1	PROVED	Thm. 4.4
N3 exposure dominates	PROVED	Thm. 4.7
Zero debt eliminates sleep phantom	PROVED	Thm. 4.8
16 ultradians/day	PROVED	Thm. 5.1
Consciousness interface		
Sensitivity parameter	MODEL	§6
Readout positivity	PROVED	Thm. 6.1
Response time law	PROVED	Thm. 6.3
Θ-coupling	PROVED	Thm. 6.4
GCIC assumption	HYPOTHESIS	conditional
Empirical interface		
Dream future-event proxy $\widehat{\mathcal{D}}_n(H)$	MODEL	§7.2
Presentiment proxy $\widehat{\Phi}_t$	MODEL	§7.2
Small-effect power envelope	MODEL	§7.3
Empirical predictions		
Deep-sleep dominance	HYPOTHESIS	Pred. 7.1
Low debt, low match rate	HYPOTHESIS	Pred. 7.2
Debt-content correlation	HYPOTHESIS	Pred. 7.3
Response time scaling	HYPOTHESIS	Pred. 7.4
Boundary reset periodicity	HYPOTHESIS	Pred. 7.5
$\sigma > 0$ for humans	HYPOTHESIS	open

10 Discussion

10.1 What This Paper Does and Does Not Claim

This paper presents a formal model, not a theory of precognition. The model shows that within a specific mathematical framework (RS), future conservation constraints create a computable cost inflation in the present. The model then shows that this inflation has a sleep-stage structure that matches the empirical coherence ordering. Whether biological neural systems can detect this inflation is an open empirical question.

The model does not require retrocausation, new particles, or violation of any established physical law. It requires only the RS axioms (which have independent motivation from information theory and discrete dynamics [20, 22]) and the standard sleep-science finding that cortical coherence decreases from waking through deep sleep.

It also does not claim that raw autobiographical narrative can be read off directly from the latent RS variables. The operational quantities in §7.2 are empirical proxies intended to preserve the model's monotonic structure under blinded, preregistered scoring.

10.2 Relationship to Existing Theories

The phantom light mechanism is compatible with, but distinct from, several existing theoretical frameworks:

- **TSVF**: Both involve future boundary conditions constraining present states. Phantom Light provides a discrete, computable version with explicit sleep-stage predictions.
- **IIT**: Both define a quantity (Φ in IIT, Φ_{mag} here) that measures the significance of temporal structure. IIT's Φ measures integrated information; Phantom Light's Φ_{mag} measures outstanding balance debt.
- **GWT**: During sleep, reduced sensory competition for the global workspace could allow phantom signals to reach broadcast—consistent with our coherence-budget model.

10.3 Limitations

1. The RS axioms are assumed, not derived in this paper. A reader who rejects the RS framework will reject the premises. The axioms are falsifiable independently [20, 23].
2. The coherence budget is a modeling choice. The qualitative predictions (ordering) are robust, but the exact ratios depend on the chosen integers.
3. The empirical proxies $\widehat{\mathcal{D}}_n(H)$ and $\widehat{\Phi}_t$ are not the latent RS variables themselves. Their adequacy must be judged by prospective predictive performance under blinded protocols.
4. The sensitivity parameter σ is a free parameter at the modeling level. Its positivity and population scale are core empirical questions.
5. The experimental predictions concern small effects in a domain with a troubled replication history. Stringent pre-registration and blinding are essential.

11 Conclusion

We have shown that within the RS framework, window neutrality creates a two-time boundary condition in which recognition events generate balance debts that the future must discharge. This balance debt inflates the present cost function, creating a "phantom light"—a computable,

bounded shadow of the future on the present cost landscape. The mechanism requires no retrocausation.

A sleep-stage hierarchy emerges: lower cortical coherence increases phantom visibility, with deep sleep (N3) dominating light sleep (N1) and REM. Zero balance debt annihilates the effect in all modes.

All mathematical claims are machine-verified (52 theorems, zero unproved placeholders). Every statement is classified as PROVED, MODEL, or HYPOTHESIS. Falsification criteria with quantitative thresholds are provided. Whether biological systems can read the phantom field is an open question. We have therefore coupled the formal structure to explicit operational proxies, a blinded prospective dream-scoring design, and small-effect study-design targets so that the empirical claims can be tested directly.

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A Lean Artifact Summary

All theorems are verified in four Lean 4 modules:

Module	**Theorems**	**Status**
`PhantomLight.lean`	17	0 sorry, 0 axiom
`PhantomSleep.lean`	22	0 sorry, 0 axiom
`SleepCycles.lean`	8	0 sorry, 0 axiom
`CayceBridge.lean`	5	0 sorry, 0 axiom
Total	**52**	**0 sorry, 0 axiom**

To verify:

```
lake build IndisputableMonolith.Consciousness.PhantomSleep
```

This transitively builds all dependencies. Source code is available at [25].